



### 3.6 Core Logic

The `domain_check()` function is the core of the Email Validation System. It validates the domain format by checking for a single dot, non-empty alphanumeric domain and TLD parts, and a valid TLD length. This modular approach makes the validation logic simple, reusable, and easy to maintain.

```
def domain_check(domain):
    # Must have exactly one dot, and only letters/digits before and after
    if domain.count('.') != 1:
        return False
    local, tld = domain.split('.')
    if not local or not tld:
        return False
    if not local.isalnum() or not tld.isalnum():
        return False # reject symbols/spaces in domain parts
    if len(tld) < 2:
        return False # TLD must be at least 2 characters
    return True
```

*Note: the system currently restricts valid emails to a fixed TLD set and alphanumeric-only local/domain parts, so addresses using hyphens, subdomains, or newer TLDs are rejected; broadening this coverage is planned as a future enhancement.*

### 3.7 Output Screenshot

